

Response to the Editor and Reviewers

Working response for author review; not cleared for submission. Submission ID: 1e58c388-d7ec-4084-b792-df20a48e3e75. The RMSE provenance and public data-cache requests remain open, as noted below. Locations refer to section titles and figure/table labels rather than page numbers from an earlier PDF.

We thank the editor and reviewers for their careful reading. We have clarified the horizon definitions, explained their practical use, and revised the presentation of the four-domain results. We also checked the stored evidence behind the numerical summaries and the alternative-baseline analysis. The corrections described below distinguish changes to the reporting from changes to experiments. No models were fitted and no predictions were recalculated during this editorial pass.

Editor

Please report results precisely, remove overstated conclusions, and explain limitations completely.

Response: We corrected a mismatch between the $\text{PM}_{2.5}$ text and its plotted LightGBM curve. That curve gives $H^*(\text{strict}) = 22$ and [27, 48]. The former value of 13 and [36, 48] came from a separate moving-average result, not the reported LightGBM curve. We corrected the text, caption, and Table 3 without changing the forecasts or the figure data.

In “Models”, the daily-load description now follows the producer: expanding training, evaluation beginning after 365 observations, and a weekly index-phase feature rather than a timestamp-derived day-of-week label. We removed claims that the observed profiles establish intrinsic domain limits or robustness across forecasting models. “Limitations” now separates positive average skill from evidence of operational benefit and makes the baseline, metric, support, model, and horizon caps explicit. The discussion of dynamic horizon selection remains a potential use, not a validated controller.

The archived four-domain seasonal-baseline analysis is reported in “Alternative-baseline sensitivity” and Table 4. Its stored comparisons were checked on exact common support; it was not rerun in this pass. **Open before submission:** the inherited primary RMSE values are not yet linked to their per-origin errors. The working manuscript does not present them as verified results.

Reviewer 1

Comment 1

In the operational definitions, use a colon before the explanations of the relaxed and strict horizons.

Response: We replaced the full stop with a colon after both labels in “Operational Predictability Horizon” (Section 3.1).

Comments 2–3

Equation (5) and the surrounding text should explicitly state that the relaxed horizon permits non-positive skill at intermediate steps.

Response: We now define

$$\mathcal{H}^+ = \{h \in \mathcal{H} : \text{Skill}(h) > 0\}$$

and

$$H^*(\text{relax}) = \max \left(\mathcal{H}^+ \cup \{0\} \right)$$

in Section 3.1. Equation (5) also defines the relaxed span $\mathcal{R}$ and the gap set $\mathcal{G} = \mathcal{R} \setminus \mathcal{H}^+$, whose elements have non-positive skill. This set may be non-empty.

The text explicitly notes that intermediate horizons with non-positive skill may fall within the relaxed reach, but are not themselves interpreted as skilful. We verified this behaviour using executable unit tests covering intermediate skill deficits and delayed onset.

Comment 4

Maintain units throughout Table 2.

Response: Table 2 now specifies $\text{PM}_{2.5}$ in $\mu\text{g m}^{-3}$, wind speed in ms^{-1} , and traffic speed in mph. For electric load, we clarify that the stored experimental target is the sum of 15-minute kW readings and that division by four expresses the aggregate daily demand in kWh, following the source metadata and the aggregation script.

Comment 5

Figures 1–3 are not correctly referenced in the text.

Response: We have added explicit textual references to the conceptual H^* schematic (Fig. 1), the rolling-origin validation protocol (Fig. 2), and the $\text{PM}_{2.5}$ skill curve (Fig. 3).

Comment 6

Evaluate baseline choices beyond persistence.

Response: We include the archived seasonal-persistence analysis, using a fixed 24-hour period for the hourly domains and a 7-day period for daily load. For each domain, we verified equality of forecast origin, target timestamp, horizon, and y_{true} between the stored LightGBM, persistence, and seasonal-persistence records. Pairwise alignment loses no records. These forecasts were generated in the earlier revision analysis; none were regenerated during the present editorial pass.

For $\text{PM}_{2.5}$, the persistence comparison on this exact common support yields $H^*(\text{strict}) = 22$ with longest contiguous positive interval $[27, 48]$, whereas seasonal persistence yields $H^*(\text{strict}) = 24$ with $[25, 48]$.

The primary LightGBM curve also gives $H^*(\text{strict}) = 22$ and $[27, 48]$, after the reporting correction explained above. The previous 13-hour value is not explained as a support effect. Equal descriptors do not imply equal samples or MAEs: the seasonal-baseline comparison uses only its matched-support rows.

Under the same matched-support analysis, load remains at $H^*(\text{strict}) = 1$, wind changes from 48 to 30, and traffic changes from 7 to 14. $H^*(\text{relax})$ remains unchanged across all four domains. We note that the hourly endpoints are exact multiples of the seasonal period, where seasonal and standard persistence coincide; the unchanged relaxed endpoint is therefore partly structural and is not interpreted as evidence of baseline invariance.

Comment 7

Explain why ARIMA(2,0,0) was selected and whether ACF was examined.

Response: We agree that the original manuscript did not document this choice adequately. An audit of our repository and Git history revealed no record that the AR(2) order was selected via ACF/PACF, AIC/BIC, grid search, auto-ARIMA, or validation splits. We therefore make no such claim.

The audit also revealed that the historical wind adapter requested a one-step ARIMA forecast at every evaluated horizon, failing to implement the stated h -step protocol. We removed the ARIMA numerical robustness claims from “Models” and the Wind and Traffic results sections rather than retaining an unsupported rationale or modifying the historical experiment post hoc.

The separate seasonal-baseline analysis tests sensitivity to the reference baseline, not robustness across forecasting models. We do not use it to justify the ARIMA order or to replace evidence of model robustness. No ARIMA fitting or order selection was performed during this revision pass.

Comments 8–9

Explain the practical utility of relaxed and strict H^ and what they add beyond fixed-horizon summaries.*

Response: A new “Operational interpretation and actionability” subsection (Section 3.2) defines $H^*(\text{relax})$ as the outer horizon at which positive baseline-relative skill is observed, and $H^*(\text{strict})$, with its interval location, as the longest contiguous positive interval.

We explain that a few fixed-horizon errors do not locate positive-skill intervals or show the gaps between them. The PM_{2.5} and traffic examples illustrate why interval locations are needed in addition to lengths. We also state that positive average skill alone does not establish deployment value: its magnitude, uncertainty, decision loss, and cost matter.

Comment 10

Could H^ dynamically decide which horizons to forecast, and could this create feedback or overfitting?*

Response: We discuss this possibility in “Discussion”. In an operational setting, H^* could inform a decision policy estimated on historical validation data, frozen for prospective deployment, and subsequently updated using only realised historical errors.

We explicitly caution that estimating H^* , selecting horizons, and evaluating performance on the same errors introduces adaptive selection bias. A dynamic controller would require prospective, prequential, or nested validation, and is not claimed to be validated here.

Reviewer 2

Comment 1

Explain the actionability of the findings more explicitly.

Response: We added the dedicated operational interpretation subsection noted above (Section 3.2), distinguishing the outer positive reach from the longest contiguous positive interval and restricting dynamic deployment claims to future, separately validated policies.

Comment 2

Figures 1 and 2 are not referenced correctly.

Response: Both figures are now explicitly introduced and discussed in the main text (Figs. 1–2).

Comment 3

In Figure 1, the curve overlaps text or annotations.

Response: The revised Figure 1 uses dedicated callout boxes and separated label placements. This is a layout correction, not a change to the horizon definitions or empirical results. We inspect it in the locally compiled manuscript; the final Overleaf rendering remains to be checked after import.

Comment 4

In Figure 2, text does not fit inside the boxes.

Response: The revised Figure 2 has larger boxes and wrapped text. We inspect the locally compiled figure for overflow and clipping. The final Overleaf rendering remains to be checked after import.

Comment 5

If possible, evaluate alternative baselines beyond persistence.

Response: We report the archived seasonal-persistence analysis detailed in our response to Reviewer 1 (Comment 6), enforcing exact common support and reporting all outcomes regardless of direction. Table 4 reports all four domains across both baselines, documenting common-origin sample counts, $H^*(\text{relax})$, $H^*(\text{strict})$, and interval locations.

Comment 6

Please cache all the data in a site you control and make the central data location clear in the paper.

Response: Not yet complete. A README, manifest, and retrieval script help reproduction but do not satisfy the request for a single author-controlled public cache. The exact four-domain inputs have been identified locally. Before submission, the permitted files must be assembled in a public location controlled by the authors, and its downloads and checksums must be verified.

“Data Availability” must then point to that location and state any redistribution restrictions. No public cache was uploaded during this editorial pass, and we do not claim that a retrieval script is one.

Comment 7

Explain the reason for $ARIMA(2,0,0)$.

Response: As noted in our response to Reviewer 1 (Comment 7), no documented order-selection rationale was preserved, and the historical wind implementation did not execute the intended h -step protocol. We removed the ARIMA robustness claims from the text and document this constraint transparently, rather than constructing a post hoc ACF/PACF justification.